Chapter 4. Classification: Logistic Regression, Softmax and SVM

Logistic regression, multiclass classification, support vector machines

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1 Odds, Logit and the Logistic Model

Motivation. For a binary label we need a model of a probability, a number in $[0,1]$, as a function of features. Modelling the log-odds as linear gives the logistic model, whose output is the sigmoid of a linear score.

For a probability p, the odds are $o = p/(1-p)$ ($p = 0.8$ gives odds 4) and

$$\operatorname{logit}(p) = \log \frac{p}{1-p} \in \mathbb{R}, \qquad \sigma(z) = \frac{1}{1+e^{-z}} = \operatorname{logit}^{-1}(z).$$

Definition 4.1: logistic regression

With augmented input $\mathbf{x} = (1, x_1, \ldots, x_d)$ and $y \in \{0,1\}$,

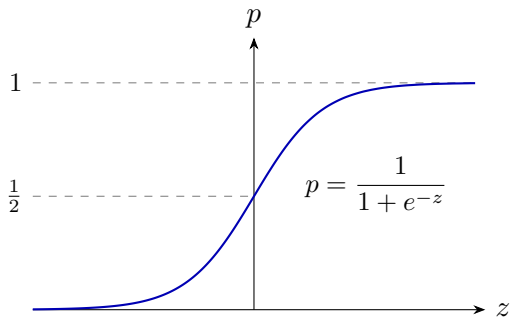
$$\log \frac{P(y=1 \mid \mathbf{x})}{P(y=0 \mid \mathbf{x})} = \mathbf{w}^\top \mathbf{x}, \qquad P(y=1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x}).$$

The score is linear and the probability transform nonlinear. Thresholding the probability at τ gives the hyperplane $\mathbf{w}^\top \mathbf{x} = \log \frac{\tau}{1-\tau}$ (at $\tau = 0.5$: $\mathbf{w}^\top \mathbf{x} = 0$).

Theorem 4.2: sigmoid identities

$\sigma'(z) = \sigma(z)(1-\sigma(z)) \le \frac{1}{4}$, $1 - \sigma(z) = \sigma(-z)$, $\frac{d}{dz}\log(1+e^z) = \sigma(z)$.

Proof. $\sigma'(z) = e^{-z}/(1+e^{-z})^2 = \sigma(z)\cdot\frac{e^{-z}}{1+e^{-z}}$ and $\frac{e^{-z}}{1+e^{-z}} = \frac{1}{e^z+1} = \sigma(-z) = 1-\sigma(z)$. The maximum of $s(1-s)$ on $[0,1]$ is $1/4$ at $s = 1/2$. ■



The logistic function, inverse of the logit $z = \log \frac{p}{1-p}$: log-odds z are mapped to probabilities p.

2 Likelihood, Gradient, Hessian and Convexity

Motivation. Logistic regression has no closed form, so we must optimize numerically. The Hessian is positive semidefinite, so the loss is convex: every local minimum is global and gradient methods are reliable.

For data $(\mathbf{x}_i, y_i)$ with $p_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$, $p(y_i \mid \mathbf{x}_i, \mathbf{w}) = p_i^{y_i}(1-p_i)^{1-y_i}$ and

$$J(\mathbf{w}) = -\sum_i \left[y_i \log p_i + (1-y_i)\log(1-p_i)\right] = \sum_i \left[\log(1+e^{z_i}) - y_i z_i\right], \quad z_i = \mathbf{w}^\top \mathbf{x}_i.$$

Theorem 4.3: logistic loss is convex

$\nabla J = \sum_i (p_i - y_i)\mathbf{x}_i = X^\top(\mathbf{p}-\mathbf{y})$ and $\nabla^2 J = \sum_i p_i(1-p_i)\mathbf{x}_i\mathbf{x}_i^\top = X^\top S X$, $S = \mathrm{diag}(p_i(1-p_i))$. Hence J is convex, and strictly convex if X has full column rank.

Proof. $\partial J/\partial z_i = \sigma(z_i) - y_i = p_i - y_i$ by the identities above, and $\partial z_i/\partial \mathbf{w} = \mathbf{x}_i$. Differentiating again, $\nabla p_i = p_i(1-p_i)\mathbf{x}_i$. For any $\mathbf{v}$, $\mathbf{v}^\top X^\top S X \mathbf{v} = \sum_i p_i(1-p_i)(\mathbf{x}_i^\top \mathbf{v})^2 \geq 0$; since $p_i(1-p_i) > 0$, the sum is positive whenever $X\mathbf{v} \neq \mathbf{0}$. ■

Example 4.4: one gradient step

$\mathbf{x} = (1,2)$, $y = 1$, $\mathbf{w} = \mathbf{0}$: $p = 1/2$, $\nabla J = (p-1)\mathbf{x} = (-1/2, -1)$. With $\eta = 0.1$, $\mathbf{w} = (0.05, 0.1)$, score 0.25, $p \approx 0.562$: the probability of the correct label increased.

3 Labels in $\{\pm 1\}$, Logistic Loss and BCE

Motivation. The same model can be written with labels $\{0,1\}$ (binary cross-entropy) or $\{\pm 1\}$ (logistic loss of the margin yf). The margin form connects logistic regression with the SVM and the other surrogate losses of this section.

With $Y \in \{\pm 1\}$ and score $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} + b$, $P(Y = +1 \mid \mathbf{x}) = \sigma(f)$ and $P(Y = -1 \mid \mathbf{x}) = \sigma(-f)$, so $-\log P(Y \mid \mathbf{x}) = \ell(Yf)$ with the **logistic loss** $\ell(z) = \log(1 + e^{-z})$. $Yf(\mathbf{x})$ is the *margin*: positive iff the sign of the score is correct.

Theorem 4.5: three names for one objective

Let $P(-1) = (1,0)$, $P(+1) = (0,1)$, $\mu(r) = (\sigma(-r), \sigma(r))$, $\tilde{Y} = (1+Y)/2 \in \{0,1\}$ and $\mathrm{BCE}(s,t) = -[t \log s + (1-t)\log(1-s)]$. For every $(\mathbf{a}, b)$ and every i,

$$D_{\mathrm{KL}}(P(Y_i) \,\|\, \mu(f(\mathbf{x}_i))) = \mathrm{BCE}\left(\sigma(f(\mathbf{x}_i)), \tilde{Y}_i\right) = \ell(Y_i f(\mathbf{x}_i)).$$

So KL minimization, BCE minimization and logistic regression are the same problem.

Proof. For a one-hot first argument, KL $= -\log$(model probability of the true class). If $Y = +1$ this is $-\log \sigma(r) = \log(1+e^{-r}) = \ell(r)$, and BCE with $t = 1$ is $-\log s$. If $Y = -1$ it is $-\log \sigma(-r) = \ell(-r)$, and BCE with $t = 0$ is $-\log(1-s) = -\log \sigma(-r)$. ■

Theorem 4.6: ℓ is convex

$\ell'(z) = \sigma(z) - 1$ and $\ell''(z) = \sigma(z)\sigma(-z) > 0$.

3.1 Separable data

Theorem 4.7: no MLE on separable data

Suppose $Y_i(\mathbf{a}^{\star\top} X_i + b^\star) > 0$ for all i. Then $\inf_{\mathbf{a},b} \sum_i \ell(Y_i(\mathbf{a}^\top X_i + b)) = 0$, the infimum is not attained, and every minimizing sequence is unbounded. With an ℓ_2 penalty $\frac{\lambda}{2}(\|\mathbf{a}\|^2 + b^2)$ a unique minimizer exists and satisfies $\|\mathbf{a}_\lambda\|^2 + b_\lambda^2 \leq 2N \log 2/\lambda$.

Proof. Let $t = \min_i Y_i(\mathbf{a}^{\star\top} X_i + b^\star) > 0$ (finitely many positive numbers). Scaling by $C > 0$ and using that ℓ is decreasing, $L(C\mathbf{a}^\star, Cb^\star) \leq N \log(1 + e^{-Ct}) \to 0$; and $L \geq 0$. Every term

is strictly positive, so $L > 0$ everywhere. If a minimizing sequence were bounded, a convergent subsequence would give a point with $L = 0$. With the penalty: $L_\lambda \geq \frac{\lambda}{2}\|(\mathbf{a}, b)\|^2$ makes the sublevel set $\{L_\lambda \leq L_\lambda(\mathbf{0})\}$ compact, so a minimizer exists (Weierstrass); strong convexity gives uniqueness; and $\frac{\lambda}{2}\|(\mathbf{a}_\lambda, b_\lambda)\|^2 \leq L_\lambda(\mathbf{0}) = N\ell(0) = N\log 2$. ■

3.2 Surrogate risk bounds

For $(X, Y) \sim P$ and classifier $f = \operatorname{sign}(g)$, counting $g = 0$ as an error, the event $\{f(X) \neq Y\}$ implies $Yg(X) \leq 0$. For a nonnegative, nonincreasing loss ϕ this gives $\phi(Yg) \geq \phi(0)$ on errors and therefore

$$P(f(X) \neq Y) \leq \frac{\mathbb{E}[\phi(Yg(X))]}{\phi(0)}.$$

Logistic: $\phi(0) = \log 2$, so $R_{\log} < \varepsilon \Rightarrow P(f(X) = Y) > 1 - \varepsilon/\log 2$. Hinge $\max(0, 1-u)$ and exponential e^{-u} have $\phi(0) = 1$. Minimizing a convex surrogate thus controls the 0–1 error, which itself cannot be optimized by gradients.

4 Multiclass Classification and Softmax

Motivation. With K classes the sigmoid generalizes to the softmax, which turns K scores into a probability vector. Its cross-entropy loss has the same clean gradient (prediction minus target) as in the binary case.

For Q classes, $W \in \mathbb{R}^{Q\times d}$, scores $s_k = \mathbf{w}_k^\top \mathbf{x}$ and

$$p_k(\mathbf{x}) = \operatorname{softmax}(\mathbf{s})_k = \frac{e^{s_k}}{\sum_r e^{s_r}}, \qquad J(W) = -\sum_{i,k} y_{ik} \log p_k(\mathbf{x}_i).$$

Scores $(2, 1, 0)$ give approximately $(0.665, 0.245, 0.090)$.

Theorem 4.8: softmax calculus

With $\operatorname{LSE}(\mathbf{s}) = \log \sum_r e^{s_r}$ and $\mathbf{p} = \operatorname{softmax}(\mathbf{s})$:

(i) $\operatorname{softmax}(\mathbf{s} + c\mathbf{1}) = \operatorname{softmax}(\mathbf{s})$, $\operatorname{LSE}(\mathbf{s} + c\mathbf{1}) = \operatorname{LSE}(\mathbf{s}) + c$; stable evaluation $\operatorname{LSE}(\mathbf{s}) = M + \log \sum e^{s_r - M}$, $M = \max s_r$.

(ii) $\nabla \operatorname{LSE} = \mathbf{p}$; Jacobian $\partial p_k/\partial s_m = p_k(\delta_{km} - p_m)$, i.e. $\nabla^2 \operatorname{LSE} = \operatorname{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top \succeq 0$.

(iii) $\nabla_{\mathbf{s}}(-\log p_y) = \mathbf{p} - \mathbf{e}_y$; for the whole data set $\nabla_W J = (P - Y)^\top X$ (rows: classes).

(iv) $\log \operatorname{softmax}(\mathbf{s}) = \mathbf{s} - \operatorname{LSE}(\mathbf{s})\mathbf{1}$ and log-softmax is idempotent: $\log \operatorname{softmax}(\log \operatorname{softmax}(\mathbf{s})) = \log \operatorname{softmax}(\mathbf{s})$.

(v) $\max_k s_k \leq \operatorname{LSE}(\mathbf{s}) \leq \max_k s_k + \log Q$.

Proof. (i) e^c cancels.

(ii) Quotient rule; $\mathbf{v}^\top(\operatorname{diag}\mathbf{p} - \mathbf{p}\mathbf{p}^\top)\mathbf{v} = \sum p_k v_k^2 - (\sum p_k v_k)^2$ is the variance of v_K with $K \sim \mathbf{p}$.

(iii) $-\log p_y = -s_y + \operatorname{LSE}(\mathbf{s})$.

(iv) With $\boldsymbol{\ell} = \log \operatorname{softmax}(\mathbf{s})$, $e^{\ell_j} = p_j$ so $\operatorname{LSE}(\boldsymbol{\ell}) = \log \sum p_j = 0$.

(v) $e^M \leq \sum e^{s_r} \leq Qe^M$. ■

Adding the same vector to all rows of W leaves the softmax unchanged, so W is identifiable only up to such shifts.

5 Support Vector Machines

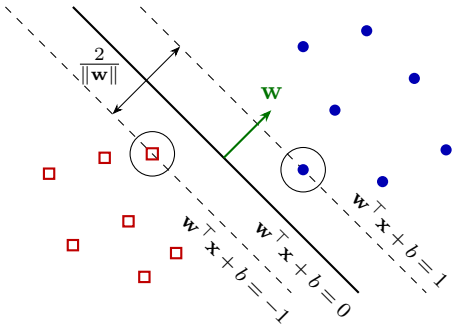
5.1 Geometry and the primal problem

Labels are $y_i \in \{\pm 1\}$, score $f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$. The distance from $\mathbf{x}_0$ to $\{\mathbf{w}^\top \mathbf{x} + b = 0\}$ is $|\mathbf{w}^\top \mathbf{x}_0 + b|/\|\mathbf{w}\|$ (move along the unit normal: $\mathbf{x}_0 - t\mathbf{w}/\|\mathbf{w}\|$ lies on the plane iff $t = (\mathbf{w}^\top \mathbf{x}_0 + b)/\|\mathbf{w}\|$). Rescaling $(\mathbf{w}, b)$ by $c > 0$ changes nothing, so for separable data normalize the closest functional margin to 1:

Definition 4.9: hard-margin SVM

$$\min_{\mathbf{w},b} \ \tfrac{1}{2}\|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1, \ i = 1, \ldots, n.$$

The support planes $\mathbf{w}^\top \mathbf{x} + b = \pm 1$ lie at distance $1/\|\mathbf{w}\|$ (the margin) from the boundary; the slab width is $2/\|\mathbf{w}\|$.



Linear SVM: the margin between the hyperplanes $\mathbf{w}^\top \mathbf{x} + b = \pm 1$ is $2/\|\mathbf{w}\|$; the circled points are support vectors.

5.2 Lagrangian, max–min inequality, weak duality

Theorem 4.10: max–min inequality

For any $F : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, $\sup_x \inf_y F(x, y) \leq \inf_y \sup_x F(x, y)$. Equality can fail: for $F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (rows x, columns y) the left side is 0 and the right side 1.

Proof. For all x, y: $\inf_{y'} F(x, y') \leq F(x, y) \leq \sup_{x'} F(x', y)$. The left end does not depend on y and the right end not on x; take $\sup_x$ on the left, then $\inf_y$ on the right. ■

With $c_i = 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)$ and $\alpha_i \geq 0$, $L(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2}\|\mathbf{w}\|^2 + \sum_i \alpha_i c_i$. Since $\sup_{\alpha \geq 0} \alpha c = 0$ if $c \leq 0$ and $+\infty$ if $c > 0$, $\sup_{\boldsymbol{\alpha} \geq \mathbf{0}} L$ equals the objective on feasible points and $+\infty$ elsewhere; hence $p^\star = \inf_{\mathbf{w},b} \sup_{\boldsymbol{\alpha}} L$. The dual is $d^\star = \sup_{\boldsymbol{\alpha}} \inf_{\mathbf{w},b} L$ and the max–min inequality gives **weak duality** $d^\star \leq p^\star$: every dual value is a lower bound on the primal optimum. For the SVM (convex, strictly feasible) strong duality $d^\star = p^\star$ holds.

5.3 The dual problem

Theorem 4.11: SVM dual

$\nabla_{\mathbf{w}} L = \mathbf{0} \Rightarrow \mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i$; finiteness of $\inf_b L$ requires $\sum_i \alpha_i y_i = 0$. The dual is

$$\max_{\boldsymbol{\alpha} \geq \mathbf{0}} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j \quad \text{s.t.} \quad \sum_i \alpha_i y_i = 0.$$

Complementary slackness $\alpha_i [1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)] = 0$: $\alpha_i > 0$ only for support vectors, and $b = y_i - \mathbf{w}^\top \mathbf{x}_i$ for any such i.

Proof. b enters L linearly with coefficient $-\sum \alpha_i y_i$; if nonzero, $\inf_b = -\infty$. Substituting $\mathbf{w}$: $\frac{1}{2}\|\mathbf{w}\|^2 - \|\mathbf{w}\|^2 + \sum \alpha_i = \sum \alpha_i - \frac{1}{2}\|\mathbf{w}\|^2$. ■

Example 4.12: two points

$\mathbf{x}_1 = (0,0)$, $y_1 = -1$; $\mathbf{x}_2 = (2,2)$, $y_2 = +1$. The constraint forces $\alpha_1 = \alpha_2 = a$ and $\|\mathbf{x}_2\|^2 = 8$, so the dual is $2a - 4a^2$, maximized at $a = 1/4$. Then $\mathbf{w} = \frac{1}{4}(2,2) = (1/2, 1/2)$, $b = -1$ (from $\mathbf{x}_1$), the boundary is $x^{(1)} + x^{(2)} = 2$, $\|\mathbf{w}\| = 1/\sqrt{2}$, margin $\sqrt{2}$ (slab width $2\sqrt{2}$). Primal value $\frac{1}{2}\|\mathbf{w}\|^2 = 1/4$ equals the dual value $2a - 4a^2 = 1/4$.

Remark 4.13: soft margin and kernels

For non-separable data add slacks: $\min \frac{1}{2}\|\mathbf{w}\|^2 + C \sum \xi_i$ s.t. $y_i f(\mathbf{x}_i) \geq 1 - \xi_i$, $\xi_i \geq 0$, equivalently hinge loss $C \sum \max(0, 1 - y_i f(\mathbf{x}_i))$ plus ℓ_2. Its dual is identical except for the box constraint $0 \leq \alpha_i \leq C$. Replacing $\mathbf{x}_i^\top \mathbf{x}_j$ by a kernel gives the kernel SVM with score $\sum_i \alpha_i y_i k(\mathbf{x}_i, \mathbf{x}) + b$. Logistic regression fits a likelihood; the SVM maximizes a geometric margin.

6 Selected exercises

Motivation. The exercises derive the second-order algorithm actually used to fit logistic regression, show how two-class softmax collapses to the logistic model, identify what the logistic loss estimates in the population, and read off the SVM margin from the dual variables.

Exercise 4.1 (Newton's method for logistic regression is IRLS)

In the setting of Section 2 ($y_i \in \{0,1\}$, $p_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$), assume X has full column rank.
(a) Show that every minimizer $\hat{\mathbf{w}}$ satisfies $X^\top(\hat{\mathbf{p}} - \mathbf{y}) = \mathbf{0}$; if the first column of X is $\mathbf{1}$, deduce $\sum_i \hat{p}_i = \sum_i y_i$.
(b) Show that the Newton step $\mathbf{w}_+ = \mathbf{w} - (\nabla^2 J)^{-1} \nabla J$ equals $\mathbf{w}_+ = (X^\top S X)^{-1} X^\top S \mathbf{z}$ with the working response $\mathbf{z} = X\mathbf{w} + S^{-1}(\mathbf{y} - \mathbf{p})$, and that $\mathbf{w}_+$ is the unique minimizer of $\sum_i S_{ii}(z_i - \mathbf{x}_i^\top \mathbf{v})^2$ over $\mathbf{v}$.
(c) Compute one Newton step from $\mathbf{w} = \mathbf{0}$ for the inputs $x \in \{-1, 0, 0, 1\}$ with labels $(0, 0, 1, 1)$, using rows $(1, x_i)$.

Solution. [Supplement]

(a) J is convex and differentiable (theorem of Section 2), so a point is a minimizer iff the gradient vanishes there: $X^\top(\hat{\mathbf{p}} - \mathbf{y}) = \mathbf{0}$. The first coordinate of this vector is $\sum_i 1 \cdot (\hat{p}_i - y_i)$.

(b) By the same theorem, $\nabla J = X^\top(\mathbf{p}-\mathbf{y})$ and $\nabla^2 J = X^\top S X$, which is positive definite because S has positive diagonal and X has full column rank. Hence

$$\begin{aligned}\mathbf{w}_+ &= (X^\top S X)^{-1}[X^\top S X\mathbf{w} - X^\top(\mathbf{p}-\mathbf{y})]\\ &= (X^\top S X)^{-1}X^\top S[X\mathbf{w} + S^{-1}(\mathbf{y}-\mathbf{p})] = (X^\top S X)^{-1}X^\top S\mathbf{z}.\end{aligned}$$

The weighted criterion is $(\mathbf{z}-X\mathbf{v})^\top S(\mathbf{z}-X\mathbf{v})$, with gradient $-2X^\top S(\mathbf{z}-X\mathbf{v})$ and Hessian $2X^\top S X \succ 0$; its unique stationary point solves $X^\top S X\mathbf{v} = X^\top S\mathbf{z}$, i.e. $\mathbf{v} = \mathbf{w}_+$. Each Newton step is a weighted least squares fit, with weights and responses recomputed from the current $\mathbf{w}$.

(c) At $\mathbf{w} = \mathbf{0}$, $p_i = \frac{1}{2}$ and $S = \frac{1}{4}\mathbf{I}_4$, so $\mathbf{z} = 4(\mathbf{y} - \frac{1}{2}\mathbf{1}) = (-2,-2,2,2)^\top$. The weights are equal and cancel: $\mathbf{w}_+ = (X^\top X)^{-1}X^\top\mathbf{z}$. Here $X^\top X = \mathrm{diag}(4,2)$ (because $\sum_i x_i = 0$ and $\sum_i x_i^2 = 2$) and $X^\top\mathbf{z} = (0,4)^\top$, so $\mathbf{w}_+ = (0,2)^\top$. The data are deliberately not separable (two points at $x=0$ carry different labels), so the pathology of the separable case does not arise. ■

Exercise 4.2 (Two-class softmax is logistic regression)

Let $Q = 2$, scores $s_k = \mathbf{w}_k^\top\mathbf{x}$, labels $y_i = y_{i1} \in \{0,1\}$, $y_{i2} = 1 - y_i$, and $J(\mathbf{w}_1,\mathbf{w}_2)$ the softmax cross-entropy.
(a) Show that $p_1(\mathbf{x}) = \sigma((\mathbf{w}_1-\mathbf{w}_2)^\top\mathbf{x})$ and $J(\mathbf{w}_1,\mathbf{w}_2) = J_{\log}(\mathbf{w}_1-\mathbf{w}_2)$, where $J_{\log}$ is the logistic loss of Section 2.
(b) Show that $\nabla_{\mathbf{w}_1}J = -\nabla_{\mathbf{w}_2}J$ and that the Hessian of J in $(\mathbf{w}_1,\mathbf{w}_2)$ is singular.
(c) For $\lambda > 0$ let $(\hat{\mathbf{w}}_1,\hat{\mathbf{w}}_2)$ minimize $J + \frac{\lambda}{2}(\|\mathbf{w}_1\|^2 + \|\mathbf{w}_2\|^2)$. Show that $\hat{\mathbf{w}}_1 = -\hat{\mathbf{w}}_2$ and that $\hat{\mathbf{w}} = \hat{\mathbf{w}}_1 - \hat{\mathbf{w}}_2$ minimizes $J_{\log}(\mathbf{w}) + \frac{\lambda}{4}\|\mathbf{w}\|^2$.

Solution. [Supplement]

(a) Dividing numerator and denominator by e^{s_1},

$$p_1 = \frac{e^{s_1}}{e^{s_1}+e^{s_2}} = \frac{1}{1+e^{-(s_1-s_2)}} = \sigma((\mathbf{w}_1-\mathbf{w}_2)^\top\mathbf{x}), \qquad p_2 = 1-p_1.$$

Hence $J = -\sum_i[y_i\log p_1(\mathbf{x}_i) + (1-y_i)\log(1-p_1(\mathbf{x}_i))]$, the logistic loss at $\mathbf{w} = \mathbf{w}_1 - \mathbf{w}_2$.

(b) By (a) and the chain rule, $\nabla_{\mathbf{w}_1}J = \nabla J_{\log}(\mathbf{w}_1-\mathbf{w}_2)$ and $\nabla_{\mathbf{w}_2}J = -\nabla J_{\log}(\mathbf{w}_1-\mathbf{w}_2)$. For any $\mathbf{v} \neq \mathbf{0}$, $t \mapsto J(\mathbf{w}_1 + t\mathbf{v}, \mathbf{w}_2 + t\mathbf{v})$ is constant, so with $\mathbf{u} = (\mathbf{v},\mathbf{v})$ the Hessian $\mathcal{H}$ satisfies $\mathbf{u}^\top\mathcal{H}\mathbf{u} = 0$. J is convex (a convex function of a linear map), so $\mathcal{H} \succeq 0$, and then $\|\mathcal{H}^{1/2}\mathbf{u}\|^2 = 0$ gives $\mathcal{H}\mathbf{u} = \mathbf{0}$ with $\mathbf{u} \neq \mathbf{0}$.

(c) The penalized objective is strictly convex (convex plus strongly convex) and coercive, so it has a unique minimizer, characterized by

$$\nabla_{\mathbf{w}_1}J + \lambda\hat{\mathbf{w}}_1 = \mathbf{0}, \qquad \nabla_{\mathbf{w}_2}J + \lambda\hat{\mathbf{w}}_2 = \mathbf{0}.$$

Adding and using (b) gives $\lambda(\hat{\mathbf{w}}_1 + \hat{\mathbf{w}}_2) = \mathbf{0}$, so $\hat{\mathbf{w}}_1 = -\hat{\mathbf{w}}_2 = \hat{\mathbf{w}}/2$. Restricted to pairs $(\mathbf{w}/2, -\mathbf{w}/2)$ the objective equals $J_{\log}(\mathbf{w}) + \frac{\lambda}{2}\cdot 2\|\mathbf{w}/2\|^2 = J_{\log}(\mathbf{w}) + \frac{\lambda}{4}\|\mathbf{w}\|^2$. The minimizer over all pairs lies on this set, so $\hat{\mathbf{w}}$ minimizes the restricted objective. The penalty thus picks the symmetric representative among all equivalent softmax parameters. ■

Exercise 4.3 (What logistic regression estimates)

Let $(X,Y) \sim P$ with $Y \in \{\pm 1\}$ and $\eta(\mathbf{x}) = P(Y = +1 \mid X = \mathbf{x}) \in (0,1)$. For measurable g let $R(g) = \mathbb{E}[\ell(Yg(X))]$ with $\ell(z) = \log(1+e^{-z})$.
(a) For $\eta \in (0,1)$ show that $C_\eta(t) = \eta\,\ell(t) + (1-\eta)\ell(-t)$ is strictly convex with unique minimizer $t^\star = \mathrm{logit}(\eta)$.

(b) Deduce that $g^\star(\mathbf{x}) = \operatorname{logit}\eta(\mathbf{x})$ minimizes R over all measurable g, and that $\operatorname{sign} g^\star$ is the Bayes classifier (predict $+1$ iff $\eta > \frac{1}{2}$).
(c) Show that $C_\eta(t^\star) = -\eta\log\eta - (1-\eta)\log(1-\eta)$, so $R(g^\star)$ is the expected binary entropy of $\eta(X)$.

Solution. [Supplement]

(a) By the theorem "ℓ is convex", $\ell'(z) = \sigma(z) - 1 = -\sigma(-z)$ and $\ell'' > 0$. Hence $\frac{d}{dt}\ell(-t) = \sigma(t)$ and

$$C'_\eta(t) = -\eta\,\sigma(-t) + (1-\eta)\sigma(t) = -\eta(1-\sigma(t)) + (1-\eta)\sigma(t) = \sigma(t) - \eta.$$

$C''_\eta(t) = \sigma'(t) > 0$, so C_η is strictly convex and its unique stationary point, $\sigma(t) = \eta$, i.e. $t = \operatorname{logit}(\eta)$, is the unique global minimizer.

(b) Conditioning on X, $\mathbb{E}[\ell(Yg(X)) \mid X] = C_{\eta(X)}(g(X))$, so $R(g) = \mathbb{E}\,C_{\eta(X)}(g(X))$. By (a), $C_{\eta(X)}(g(X)) \geq C_{\eta(X)}(g^\star(X))$ pointwise; taking expectations gives $R(g) \geq R(g^\star)$, where $R(g^\star) \leq C_\eta(0) = \log 2$ is finite and $g^\star$ is measurable as a composition of measurable maps. Finally $\operatorname{logit}\eta > 0$ iff $\eta > \frac{1}{2}$.

(c) From $\sigma(t^\star) = \eta$: $\ell(t^\star) = -\log\sigma(t^\star) = -\log\eta$ and $\ell(-t^\star) = -\log\sigma(-t^\star) = -\log(1-\eta)$. Substituting into C_η gives the entropy. So minimizing the logistic loss estimates the conditional log-odds, not just the decision; this is why its outputs can be read as probabilities. ■

Exercise 4.4 (The margin from the dual variables)

Let the data be linearly separable with both labels present, let $(\mathbf{w}^\star, b^\star)$ solve the hard-margin primal and $\boldsymbol{\alpha}^\star$ the dual, with the KKT conditions of the dual theorem in force.
(a) Show that $\mathbf{w}^\star$ is unique, $\mathbf{w}^\star \neq \mathbf{0}$, and $b^\star$ is uniquely determined.
(b) Show that $\|\mathbf{w}^\star\|^2 = \sum_i \alpha_i^\star$; hence the margin is $\left(\sum_i \alpha_i^\star\right)^{-1/2}$ and the primal and dual values coincide.
(c) Check (b) on the two-point example.

Solution. [Supplement]

(a) *Step 1 (uniqueness of $\mathbf{w}^\star$).* The feasible set F is an intersection of half-spaces in $(\mathbf{w}, b)$, hence convex. Suppose $(\mathbf{w}_1, b_1), (\mathbf{w}_2, b_2) \in F$ are both optimal with value $p^\star$ and $\mathbf{w}_1 \neq \mathbf{w}_2$. Their midpoint is feasible and, by the parallelogram identity,

$$\tfrac{1}{2}\left\|\tfrac{\mathbf{w}_1+\mathbf{w}_2}{2}\right\|^2 = \tfrac{1}{4}\|\mathbf{w}_1\|^2 + \tfrac{1}{4}\|\mathbf{w}_2\|^2 - \tfrac{1}{8}\|\mathbf{w}_1 - \mathbf{w}_2\|^2 < p^\star,$$

a contradiction.

Step 2 ($\mathbf{w}^\star \neq \mathbf{0}$). With $\mathbf{w} = \mathbf{0}$ the constraints read $y_i b \geq 1$; a point with $y_i = +1$ forces $b \geq 1$ and one with $y_i = -1$ forces $b \leq -1$. So $\mathbf{w}^\star \neq \mathbf{0}$.

Step 3 ($b^\star$). Since $\mathbf{w}^\star = \sum_i \alpha_i^\star y_i \mathbf{x}_i \neq \mathbf{0}$, some $\alpha_i^\star > 0$. Complementary slackness gives $y_i(\mathbf{w}^{\star\top}\mathbf{x}_i + b^\star) = 1$, and $y_i^2 = 1$ gives $b^\star = y_i - \mathbf{w}^{\star\top}\mathbf{x}_i$, a function of $\mathbf{w}^\star$ alone.

(b) Summing complementary slackness over i and using $\sum_i \alpha_i^\star y_i = 0$,

$$\sum_i \alpha_i^\star = \sum_i \alpha_i^\star y_i(\mathbf{w}^{\star\top}\mathbf{x}_i + b^\star) = \mathbf{w}^{\star\top}\sum_i \alpha_i^\star y_i \mathbf{x}_i + b^\star \sum_i \alpha_i^\star y_i = \|\mathbf{w}^\star\|^2.$$

The margin is $1/\|\mathbf{w}^\star\|$. The primal value is $\frac{1}{2}\|\mathbf{w}^\star\|^2$, and the dual value is $\sum_i \alpha_i^\star - \frac{1}{2}\|\mathbf{w}^\star\|^2 = \frac{1}{2}\|\mathbf{w}^\star\|^2$.

(c) There $\alpha_1 = \alpha_2 = \frac{1}{4}$, so $\sum_i \alpha_i = \frac{1}{2} = \|(1/2, 1/2)\|^2$ and the margin is $(1/2)^{-1/2} = \sqrt{2}$, as found directly. ■